



Enhancing a fuzzy system through computational intelligence-based feature selection for decision-making in cloud computing environments[☆]

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ABSTRACT

The present study addresses the challenges associated with flexible resource management in Cloud Computing, focusing on optimizing resource utilization. Specifically, we investigate the problem of dynamic server consolidation, leveraging the capabilities of Fuzzy Logic to address uncertainties and imprecision inherent in cloud environments. Our research aims to enhance decision-making support by extending a fuzzy system to determine the utilization level of physical machines within a Cloud Computing environment. This extension involves refining the inference model through a Feature Selection process, employing computational intelligence techniques inherent in the Fuzzy Rule-Based Classification System. Comparative evaluations of classification algorithms highlight the notable accuracy of FURIA and IVTURS, which are emerging as close alternatives. Additionally, we assess the proposed extension by comparing it with the definitions utilized in the Int-FLBCC system currently under development by our research group. The results underscore the promising nature of our system extension, offering valuable perspectives for decision-making processes within the Cloud Computing environment.

1. Introduction

Cloud Computing (CC) is a computational paradigm for ubiquitous and on-demand resource sharing on the network, categorized based on service models, including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) and Function as a Service (FaaS) [1].

The global public market for CC prospects above US\$ 679 billion by 2024.¹ This estimation encompasses various services delivered by public CC providers, including business processes, platforms, infrastructure, software, management, security, and services such as storage, bandwidth, or CPU cycles.

In this context associated with CC environments, efficient resource management emerges as a critical demand. In particular, energy efficiency holds high significance due to its impact on various fronts, including the maintenance of satisfactory Service Level Agreements

(SLAs) and Quality of Service (QoS) [2,3]. Minimizing energy consumption while satisfying QoS constraints is a complex and central research task to dynamic Virtual Machine (VM) consolidation, characterized as an NP-Hard problem [4].

VM consolidation, more specifically, involves identifying underloaded/overloaded physical hosts and selecting VMs for migration to the most suitable host among these [5]. This strategy can be highly complex, considering the potential sudden and dynamic changes in the CC environment [6].

Fuzzy Logic (FL) [7] is frequently employed to assist in decision-making processes, addressing uncertainties and inaccuracies in the variables involved in VM consolidation.

In 1975, Sambuc [8] presented the concept of an Interval-valued Fuzzy Set (IvFS), establishing relevant studies, taking into account the uncertainty linked to a precise interval-valued membership function [9].

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¹ <https://www.statista.com/statistics/203578/global-forecast-of-cloud-computing-services-growth>

The ability to make decisions under uncertainties and tolerate imprecision in control systems motivates the development of CC management applications, based on the Interval-valued Fuzzy Logic (IvFL). More recently, this logical approach has contributed to solving complex problems by reasoning models, deduction, and calculating imperfect information, integrating techniques from Artificial Intelligence such as Machine Learning and Neural Networks [10].

Similarly, reliability allocation faces difficulties in other complex system domains, such as reliability engineering, due to epistemic uncertainty and imprecise failure data. These comments use approaches like fuzzy logic to handle the imprecision, offering an alternative to traditional methods that require precise data, as proposed in [11].

Allocation and optimization problems are central to efficient resource management in complex systems. In Cloud Computing, these challenges manifest in the context of VM consolidation, while in systems engineering, they encompass tasks such as optimal reliability allocation. Addressing these problems, a variety of computational intelligence algorithms have been employed. In [12], for instance, the Jaya algorithm was applied to optimize reliability allocation in complex systems, such as the oxygen supply system of a spacecraft, aiming to minimize overall system costs without compromising reliability.

This work constitutes an expansion and deepening of the previous research in [13], which focused on the application of Feature Selection on data from a simulated CC environment, restricted to VM allocation and selection policies adopted by the Int-FLBCC [14] approach, for achieving the virtual machine consolidation. The most relevant variables were identified, and new membership functions and rule bases were developed through Fuzzy Rule-Based Classification Systems.

Expanding upon these investigations, we explore a variety of combinations between different allocation and selection policies available in the simulated environment. Feature Selection is performed for each evaluated configuration to identify the most significant variables. By selecting these variables, new membership functions and rule bases are dynamically generated by Fuzzy Rule-based Classification Systems.

Thus, our proposal carries out a Feature Selection process, refining inference models and leveraging computational intelligence techniques inherent in Fuzzy Rule-Based Classification Systems, prospecting a wide range of research outcomes on energy-aware computational resource management.

The paper is structured as follows. Section 2 provides the contextual foundations of the work. Related work is reviewed in Section 3. Methodological details, including Exploratory Data Analysis and dataset parameters, are discussed in Section 4. The Feature Selection results are presented in Section 5. Finally, Section 6 outlines conclusions and future work.

2. Preliminaries

In this section, we report the foundation studies and basic concepts, starting with the challenges of CC sustainability and introducing Fuzzy Rule-Based Classification Systems. Additionally, we present the Int-FLBCC system [15].

2.1. Cloud computing sustainability challenges

The Cloud Computing operational model enables dynamic resource allocation based on the demand placed on the computational environment [16]. This elasticity of resources facilitates the provisioning of high-performance computational environments while optimizing equipment investments for end-users, thereby aligning costs with resource requests [17].

Since 2014, data centers in the United States have consumed an estimated 70 billion kilowatt-hours (kWh), accounting for approximately 1.8% of the total USA electricity consumption, as reported by the Natural Resources Defense Council (NRDC) [18]. This significant energy usage raises concerns regarding its alignment with several

World Health Organization Sustainable Development Goals.² Particularly, it conflicts with goals such as (i) Affordable and Clean Energy, (ii) Responsible Consumption and Production, and (iii) Climate Action.

Moreover, energy consumption in the USA data centers increased by approximately 4% from 2014 to 2023, reaching an estimated 73 billion kWh in 2023.

This substantial increase underscores the urgency for research aimed at energy optimization. Consequently, leading global companies such as Google,³ Microsoft,⁴ and Amazon⁵ are actively pursuing sustainability goals by transitioning to renewable energy sources and investing in on-site green energy generation.

Efficient resource management in Cloud Computing can significantly benefit from VM consolidation. This involves identifying overloaded and underutilized physical machines, selecting VM migration strategies, and reallocating resources to alternative physical machines.

2.2. CloudSim architecture

The strategies and algorithms assessments in CC environments require simulation-effective tools. In this context, CloudSim [19] emerges as a widely employed and esteemed simulation framework, providing researchers with a modular platform for analyzing policies, algorithms, and strategies [20].

The CloudSim architecture design⁶ enables a comprehensive and adaptable simulation of CC environments. The key entities interoperate across abstraction layers:

- **CloudSim Core simulation engine layer** supports simulation and modeling of cloud surroundings;
- **CloudSim layer** provides management interfaces for VM, memory, storage, bandwidth, and also the provisioning of hosts to VM, managing application execution, and monitoring system state;
- **User code layer** enables users to customize the environment's characteristics.

This architecture encompasses crucial modules, such as resource provisioning and scheduling policies. Extensibility is inherent to CloudSim, allowing the addition of modules and policies to meet specific research or simulation requirements. The modular and scalable nature of CloudSim facilitates its application in diverse scenarios, contributing to detailed simulations in cloud computing environments.

Furthermore, compared to physical environments, the possibility of experiment repetitions in a controlled manner, improving the reproducibility of the tests, is one of the main advantages arising from the use of simulation tools, such as CloudSim, integrating the synergistic variation of the different conditions for more complex evaluations.

2.3. Fuzzy rule-based classification systems

Fuzzy Rule-Based Classification Systems (FRBCSs) consist of potent tools commonly employed to address classification problems [21]. Renowned for their high classification accuracy and ability to provide interpretable models through linguistic labels [22,23], FRBCSs have two main components: (i) the Knowledge Base, comprising a specialized Rule Base and Database adapted to a specific classification problem; and the (ii) Fuzzy Reasoning Method, which applies fuzzy logic to manage data uncertainty and to assign class labels effectively [24].

FRBCS design encompasses rigorous supervised learning, initiated with a set of correctly classified training data. The primary objective is to formulate a Classification System capable of minimizing errors in

² <https://brasil.un.org/pt-br/sdgs>

³ <https://environment.google>

⁴ <https://www.microsoft.com/en-us/environment/energy>

⁵ <https://aws.amazon.com/about-aws/sustainable-energy>

⁶ <http://www.cloudbus.org/cloudsim>

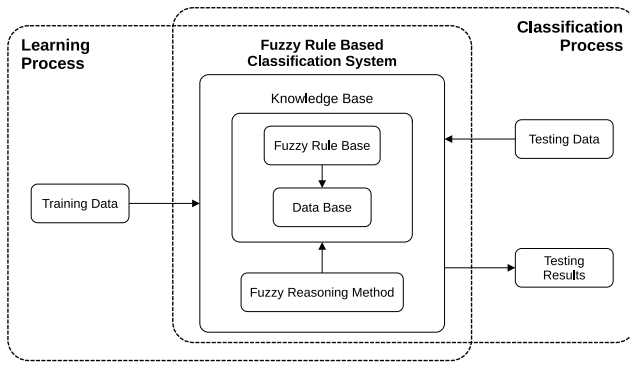


Fig. 1. Fuzzy Rule-based Classification Process from Training to Testing [24].

assigning class labels to novel instances. The system's performance is evaluated comprehensively on test data, approximating the FRBCS real error. This structured approach underscores the systematic construction and performance assessment intrinsic to the FRBCS, substantiating its utility in intricate classification scenarios. Fig. 1 shows this process.

Conducting this study, the following examples of FRBCSs are summarized:

- **Fuzzy Association Rule-based Classification model for High Dimensional problems (FARC-HD)** [25] is a classifier designed for high-dimensional problems, aiming to produce accurate, interpretable models with low computational cost. The method is structured into three main steps: (1) extraction of concise fuzzy association rules using a limited-depth search strategy to ensure simplicity and interpretability; (2) pre-selection of the most relevant rules based on a quality criterion inspired by Subgroup Discovery to reduce computational complexity in the following stage; (3) genetic-based rule selection and fine-tuning of the linguistic terms, allowing adjustments that preserve the original shapes of the membership functions to enhance classification performance. The method employs the Certainty Factor as the rule weight and adopts a fuzzy reasoning approach based on weighted voting. The initial fuzzy partitions are uniform, with five triangular membership functions, adding a default rule for the majority class.
- **Chi et al. algorithm** [26] employs a grid-based approach with fixed fuzzy partitions for the attributes. For each grid cell, the algorithm identifies the training instance with the highest membership degree and assigns its class as the consequent of the rule. Using three triangular fuzzy sets per attribute, the reasoning process applies a product t-norm. The Chi algorithm tends to generate large rule sets, with rules containing multiple conditions, which may impact the interpretability.
- **Fuzzy Unordered Rule Induction Algorithm (FURIA)** [27] Furia is a fuzzy extension of the well-known RIPPER [28] algorithm. Unlike RIPPER, which produces ordered rule lists, FURIA learns unordered rules, typically using a one-vs-rest scheme for multiclass classification. The rules are fuzzy, with boundaries represented by intervals, allowing for more flexible modeling of decision boundaries. When no rule covers an instance, the algorithm applies rule stretching to locally generalize existing rules. Based on the highest support, the final decision computes the sum of coverage degrees weighted by a Certainty Factor.
- **Interval-Valued fuzzy reasoning method with Tuning and Rule Selection (IVTURS)** [29] extends FARC-HD by incorporating IvFS, which provides a more robust representation of uncertainty and ignorance in linguistic terms. Its main innovation is the Interval-valued Fuzzy Reasoning Method, which uses Interval-Restricted Equivalence Functions (IV-REF) to compute

the matching degree between instances and rule antecedents. The initial rule base is generated via FARC-HD, with linguistic labels modeled as IvFS. The system is then optimized using a genetic algorithm that simultaneously tunes the parameters of the IV-REF and performs rule selection, aiming to reduce model complexity while improving classification accuracy.

Table 1 summarizes the main characteristics of the fuzzy rule-based classification systems (FRBCS) discussed in this study, highlighting their objectives, underlying techniques, performance, interpretability, and main limitations.

The algorithms discussed and employed throughout this work have been utilized in previous studies conducted by our research group. These algorithms are implemented and validated in the KEEL software tool [30], which is well-established and recognized by the international scientific community.

The adoption of fuzzy strategies, in particular the IvFL approach based on the IVTURS methodology, aligns with the ongoing research directions of our group. These strategies constitute a robust framework for integrating our work within the large field of Computational Intelligence.

2.4. Int-FLBCC

The Interval Fuzzy Load Balancing for Cloud Computing (Int-FLBCC) [14] is a computational decision-making system considering a flexible approach for the VM consolidation, founded by concepts of Interval-valued Fuzzy Sets (IvFSs). The Int-FLBCC fuzzy system design addresses the uncertainty and imprecision related to the dynamism and fluctuations of resources in physical machines in the infrastructure of Cloud Computing.

The system tackles issues in dynamic server consolidation to optimize energy consumption, reduce SLA violations, and minimize the number of VM migrations. Int-FLBCC identifies the computational power, communication cost, and RAM utilization level, which are normalized and undergo all stages of an interval-valued fuzzy inference system. Additionally, it involves gathering information about the runtime resources of the CC environment and processing a fuzzy rule-based system to return a list of the corresponding physical machines' utilization level hosts, and then delivering it to the scheduler. See Fig. 2 illustrating this process.

3. Related work

This section reports the main projects proposing fuzzy strategies in the stages of dynamic server consolidation in Cloud Computing. The selected papers resulted from a Systematic Literature Review (SLR) realized in [32]. We identify the primary objectives of each work, the methodological approach system, the resources considered for decision-making, and the algorithm employed.

In the work by [33], the authors employ a combination of Machine Learning (ML) and Type-2 Fuzzy Logic (T2LF) in the proposed Clustering-based Multiple Objective Dynamic Load Balancing (CMODLB) technique. The aim is to enhance resource utilization, reduce task completion and transmission times, improve energy consumption, and achieve load balancing across PMs. The ML strategy is responsible for clustering VMs based on resource load using a Bayesian optimization of the K-means algorithm named BOEK-means. Additionally, the system features a task scheduler on VMs supported by the TOPSIS-PSO algorithm.

In [34], the fuzzy approach called SARSA (State, Action, Reward, State, Action) Reinforcement Learning (FSRL) combines Reinforcement Learning (RL) techniques with Fuzzy Logic to optimize the allocation (and) or relocation of VMs to reduce energy consumption. During the VM management running on each host, the SARSA RL algorithm guides VM allocation by monitoring the dynamic behavior of each host in the

Table 1
Summary of main characteristics of selected FRBCS.

FRBCS	Main Objective	Techniques Used	Performance	Interpretability	Main Limitations
Chi	Proposing a simple fuzzy rule-based classification system using association rules	Discretization of attributes; fuzzy association rules based on support and confidence	Solid foundation but limited performance in complex datasets	Very high (simple and linguistic rules)	Low accuracy; overfitting due to large number of rules; no rule selection or tuning; poor handling of high-dimensional data
FARC-HD	Handling high-dimensional problems with optimized fuzzy rules	Fuzzy rule mining; genetic algorithms for rule selection and lateral tuning	High accuracy and generalization	Moderate balance (reduced with tuning)	High computational cost; sensitive to genetic parameter settings; interpretability decreases with tuning
FURIA	Enhancing flexibility and robustness in fuzzy rule induction	Fuzzy version of RIPPER; unordered rules; adaptive fuzzy intervals	Very high accuracy and robustness; strong generalization	Low (complex and unordered rule structure)	Difficult linguistic analysis; reduced transparency; moderate computational cost
IVTURS	The better representing uncertainty using interval-valued fuzzy sets (IvFS)	IvFS; heuristic rule selection and tuning; linguistic rule expressions	Excellent robustness to noise and uncertainty; strong generalization	Good (linguistic rules with interval representation)	High computational complexity; challenging IvFS implementation; sensitive interval modeling

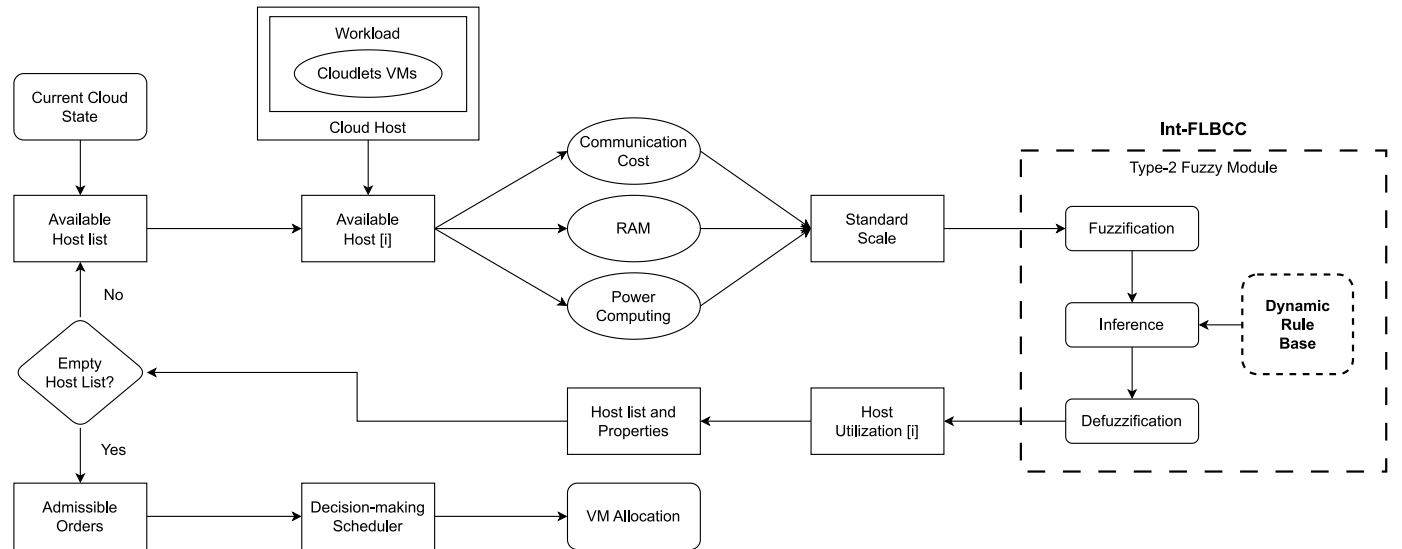


Fig. 2. Overview of Int-FLBCC Features [31].

environment. The fuzzy system supports SARSA by creating fuzzy sets and rules that narrow the set of active hosts, thereby minimizing the search time for VM allocations.

The study by [35] utilizes the Fuzzy Emperor Penguin Optimization (Fuzzy-EPO) algorithm, which proposes the reallocation of overloaded VMs to underutilized PMs, reducing the time of migration in a CC environment. The system also incorporates the Energy Efficient Brownout Approach (EEBA), a brownout mechanism based on the Placement and Consolidation (PCO) algorithm [36]. This mechanism serves as a strategy to handle overload situations by temporarily reducing allocated resources, aiming to prevent a complete service interruption during periods of high demand. The system considers CPU, memory, storage, and bandwidth to evaluate the state of PMs.

In [37], the authors introduce the Fuzzy Best-Fit-Decreasing (fuzzyBFD), a multi-objective approach with the best-decreasing fit for the VM allocation problem. The system examines the existence of overloaded physical machines and adds candidate VMs for migration to a migration list. For underutilized PMs, the system checks the feasibility of migrating all their VMs and adds them to the list. After analyzing the PMs, the system seeks the new allocation for each VM in the list.

The results in [38] present the Performance and Resource-aware Virtual Machine Selection using Fuzzy (PRSF), considering a VM selection policy to maximize CPU resource utilization and minimize VM migration. The system reports the historical CPU usage and calculates the threshold value for the Membership Function. The resource manager invokes VM relocation based on overload detection, moderating resource usage. PRSF identifies the optimal VM for migration to ensure efficient resource utilization while minimizing SLA violations. By adding the selected VM to a migration list, further relocations are demanded until the PM's predefined resource limits are achieved.

In [39], a multifunctional approach combining Fuzzy Logic and Granular Computing (GC) [40] addresses the issue of excessive VM migrations, aiming to minimize energy consumption and maximize QoS stability. The classification rules and appropriate VM identification consider request characteristics.

3.1. Analysis and discussion

The SLR analysis reveals the dynamic VM consolidation, emerging as an effective strategy to enhance efficient energy in CC, based on four steps: (i) Overloaded PM Identification; (ii) VM Selection; (iii) Detection of Underutilized PMs; and (iv) Optimized VM Allocation. The

Table 2
Summarized analysis of selected papers in the SRL process.

Strategy	Goal	LA	Variables	Tools
Int-FLBCC	Optimizing energy consumption, reducing SLA violations, and minimizing the number of VM migrations	IvFL	Computational Power; Communication Cost; Memory	■ □
CMODLB	Improving resource utilization, load balancing and energy consumption, reducing task and transmission times	ML T2FL	CPU; Memory; Load Balance in PM	■
FSRL	Reduction of energy usage and resource wastage	FRL	CPU; Memory	■ ▲
Fuzzy-EPO	Minimizing VM migration time and reducing energy consumption	FL	CPU; Memory; Storage; Bandwidth	■
fuzzyBFD	Improving energy consumption and resource utilization	FL	CPU; Memory; Energy; Storage	■
PRSF	Optimizing VM migration for SLA assurance	FL	CPU; Memory	■
GRC model:	Minimizing energy consumption and maximizing QoS stability	GC	CPU; Core Number; Memory; Storage; Service Time; Request Number	■

Logic Approach FL: Fuzzy Logic; IvFL: Interval-valued Fuzzy Logic; ML: Machine Learning; T2FL: Type-2 Fuzzy Logic; FRL: Fuzzy Reinforcement Learning; GC: Granular Computing; ■ CloudSim; □ Juzzy; ▲ Matlab.

underlying goal is to achieve dynamic consolidation of VMs, aiming to optimize the trade-off between performance and energy efficiency.

Table 2 presents an analysis of proposals for resource management in CC, applying techniques on dynamic server consolidation, and highlighting their significance. The description emphasized objectives, logical approach, variables, and prospected tools.

This analysis reveals a pronounced focus on the optimization and minimization of energy consumption in CC environments as a crucial theme to meet the demand for consumption reduction.

Two works stand out for the integration of FL with ML. In [33], ML uses cluster VMs based on resource load. Meanwhile, in [34], FL combines with RL, optimizes the allocation and/or relocation of VMs. Integrating ML techniques allows FL systems to learn from data, adapting automatically to changes and complexities in the problem. Both approaches enable the improvement of fuzzy rules based on the characteristics and relationships present in the data. The other works do not address ML techniques.

Regarding the logical approach, only [14,33] describe works based on the multivalued extension of FL, specifically IvFL and T2FL, respectively. The strategy in [39] considers Granular Computing. The other works adopt Type-1 Fuzzy Logic.

The assessment of computational power and memory usage is stated as a consensus among the works regarding the variables considered by Fuzzy Systems. In [33,34,37,38], and [39], the authors optimize the allocation or minimize the migration time of VMs. However, they do not account for communication costs, which can lead to bottlenecks in the system when congested, and consequently may lead to unsatisfactory performance.

Another convergence states CloudSim as a simulation tool for CC environments. Thus, our proposed feature selection also considers variables available by CloudSim.

4. Feature selection methodology

This section explores the evolving process of Feature Selection and evaluates the classification performance of different algorithms for VM selection in dynamic VM consolidation to improve energy efficiency in cloud data centers.

Feature Selection methods provide us a way of reducing computation time, improving prediction performance, and data understanding in machine learning or pattern recognition applications [41]. To conduct a Feature Selection in this research, we use a dataset derived from CloudSim executions, incorporating a combination of allocation policies (AP) and promoting decision-making based on statistical analysis of historical data for calculating host utilization.

Since fixed values of utilization thresholds are unsuitable for an environment with dynamic and unpredicted workloads, sharing a physical resource in different types of applications seems reasonable. The system should be able to automatically adjust its behavior depending on the exhibit application workload patterns. The four elected allocation algorithms incorporate three different strategies:

- **Inter Quartile Range (IQR)**, containing the second and third quartiles, or the middle half of a selected data set, telling the spread of the middle half of the data distribution;
- **Local Regression (LR)** [42] and **Local Regression Robust (LRR)** [43], exploring regression based on algorithms for future estimation of CPU utilization;
- **Median Absolute Deviation (MAD)**, based on mean, median, and standard deviation, focuses on terminal/outlier values and then decides which VM category should migrate.

The selection policies (SP) include Maximum Correlation (MC), Minimum Migration Time (MMT), Minimum Utilization (MU), and Random Selection (RS), considering a uniformly distributed discrete random variable.

The dataset used in the experiments was obtained through a simulated cloud computing environment over a 24-hour simulation period, with resource utilization data collected every 300 s, using PlanetLab⁷ workload traces, executed in CloudSim. Each record captures system metrics such as CPU usage, memory occupancy, bandwidth, storage, energy consumption, and MIPS (Millions of Instructions per Second).

PlanetLab workload traces are widely adopted in cloud computing research for benchmarking scheduling and allocation policies, as emphasized in the comprehensive survey in [44].

One dataset was generated per AP×SP combination, resulting in 16 allocation–selection scenarios. Feature selection was applied individually, yielding tailored attribute subsets.

About 320,000 instances were collected, with each dataset averaging 20,000 samples. Each sample was labeled as underutilized, regular, or overutilized, based on CloudSim internal metrics [13].

Regarding the dataset, the Feature Selection considers the Sequential Forward Selection (SFS) [45] technique to eliminate highly correlated variables. The aim is twofold: the former enhances overall classification performance and reduces computational effort, and the latter involves processing data. The selection process involves iteratively combining variables to achieve optimal subset configurations based on evaluation criteria. Fig. 3 illustrates this process.

⁷ <https://planetlab.cs.princeton.edu>

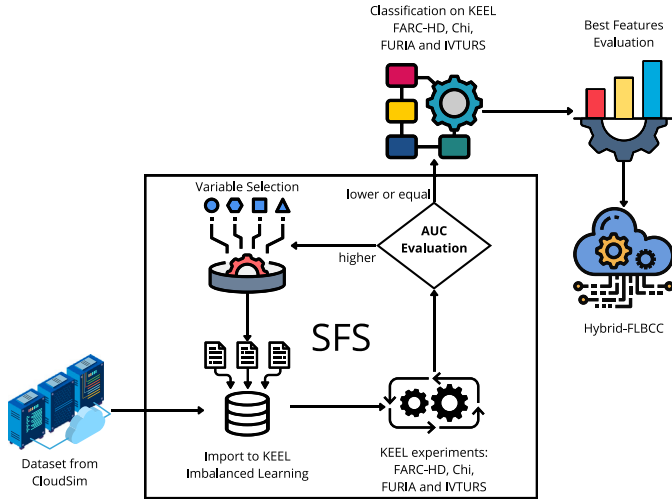


Fig. 3. SFS iteration flow.

Based on the configurations with the best classification performance results, we can define new membership functions and rule bases to provide a more straightforward interpretation for the selected scenario, simplifying the fuzzy system implementation and potentially saving computational resources.

As applied in our previous work [13], for this study, we also adopted the Area Under the ROC Curve (AUC) [46] as the evaluation criteria metric, which strategically enhances the assessment of the classification performance, providing a comprehensive measure of the trade-off between true and false positives rates across different classification thresholds.

The AUC metric is particularly well-suited for our objectives, as it offers a holistic evaluation of the model's discriminatory capacity, considering various decision thresholds and encompassing a broader understanding of its predictive capabilities.

To ensure robustness and minimize bias, a 10-fold cross-validation strategy was employed: the dataset was partitioned into ten equal-sized folds, with each fold used once for testing and the remaining nine for training. This process was repeated ten times, and the AUC was calculated individually for each fold. The final results report the mean AUC across all folds.

This work implements the Sequential Forward Selection technique using the KEEL software tool [30], designed to implement evolutionary algorithms and soft computing techniques, focusing on data mining problems. SFS is executed with FARC-HD [47], Chi [48], FURIA [28], and IVTURS [49] algorithms (see Section 2.3), all utilizing default parameters.

In this study, we aimed to design a generalizable data analysis model, capable of supporting decision-making across a variety of cloud computing scenarios. For this reason, standardized parameter settings were adopted to ensure broad applicability and reproducibility of the results, rather than tailoring the configuration to a specific cloud infrastructure.

For each combination of allocation and selection policies, we executed the four algorithms to achieve the best Feature Selection for each scenario. The results were then compared with the metrics of features considered in the Int-FLBCC system [50]: CPU, Memory, and Bandwidth regarding fuzzy correlation analysis.

5. Experimental results

Decision-making systems enhance the quality of results, robustness, and adaptability of decisions by stressing scenarios related to numerous variables, facilitating a more comprehensive and informed approach to

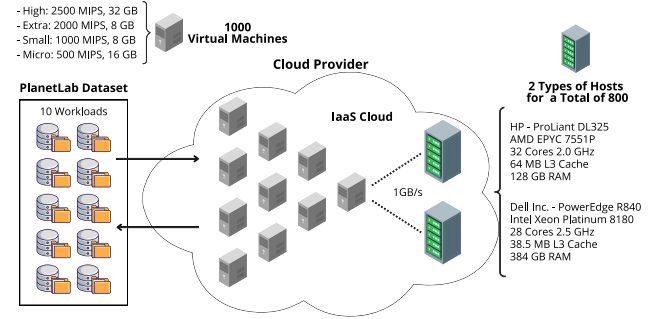


Fig. 4. Specification of the IaaS Cloud Environment.

resolving complex problems. The promoted solutions by simulation experiments increase the learning about the system's capacity to capture complex relationships, better adapt to diverse contexts, identify subtle patterns, and mitigate uncertainties, thereby enabling more accurate and resilient decision-making processes.

This section provides all information about the classification performance and optimal variable combinations for each considered algorithm, summarized in Table 4, Table 5, and Table 6.

Unlike the initial implementation of Int-FLBCC, which employs only the "CPU", "RAM", and "BW" variables for decision-making across all policy combination scenarios, the Feature Selection performed reveals significantly different configurations depending on the scenario and the considered algorithm. In some cases, the number of variables exceeded the fixed quantity in the system.

Based on the configurations with the best classification performance results, we can define new membership functions and rule bases to provide a more straightforward interpretation for the selected scenario, simplifying the fuzzy system implementation and potentially saving computational resources.

The comparative evaluation of classification algorithms highlighted differences in their performance and approaches. FURIA exhibited notable precision with consistently high AUC values, solidifying its effectiveness. IVTURS showed comparable performance, positioning it as a promising alternative.

In addition to FARC-HD and Chi, FURIA employs Type-1 Fuzzy Logic. IVTURS extends the multi-valued fuzzy approach with IvFSs, enhancing its ability to handle complex uncertainties in server consolidation case studies.

5.1. Cloud environment setup

The CC environment and the evaluation of the fuzzy module for detecting the hosts' load levels were conducted using the CloudSim simulator in version 3.0. The IaaS-type cloud environment defined for the case studies consisted of a large-scale data center comprising 800 heterogeneous hosts, with two configuration types as shown in Table 3 and Fig. 4.

The CPU frequency of the hosts was specified in MIPS (Millions of Instructions Per Second). Half of the hosts were modeled as ProLiant DL325 Gen10 servers, with each core delivering 4721 MIPS, whereas the other half were modeled as PowerEdge R840 servers, with each core delivering 4520 MIPS. Each host was configured with a network bandwidth of 1 GB/s.

The VM types corresponded to the following instance configurations: (1) Medium-High CPU Instance (2500 MIPS, 32 GB RAM); (2) Extra Large Instance (2000 MIPS, 8 GB RAM); (3) Small Instance (1000 MIPS, 8 GB RAM); and (4) Micro Instance (500 MIPS, 16 GB RAM).

Table 3

Host Configuration in the Simulated Environment.

Vendor	Model	CPU Name	CPU Characteristics	Memory
HP	ProLiant DL325 Gen10	(1)	32-Core, 2.0 GHz, 64 MB L3 Cache	128 GB
Dell	PowerEdge R840	(2)	28-Core, 2.50 GHz, 38.5 MB L3 Cache	384 GB

(1) AMD EPYC 7551P 2.0 GHz

(2) Intel Xeon Platinum 8180 2.50 GHz

Table 4

Global Results of Feature Selection Simulation for IQR and LR.

AP	SP	FRBCS	Variables							AUC	stddev
IQR	MC	FARC-HD	cpu	bw	storage	–	–	–	–	0.9102	0.0087
		FURIA	cpu	energy	mips	–	–	–	–	0.9851	0.0050
		IVTURS	cpu	bw	mips	–	–	–	–	0.8812	0.0311
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8524	0.1322
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9209	0.0144
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8419	0.0095
	MMT	FARC-HD	cpu	bw	–	–	–	–	–	0.9249	0.0088
		FURIA	cpu	energy	mem	storage	–	–	–	0.9954	0.0021
		IVTURS	cpu	bw	mips	mem	–	–	–	0.9301	0.0109
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9218	0.0152
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9292	0.0144
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8039	0.0099
LR	MU	FURIA	mips	energy	cpu	–	–	–	–	0.9885	0.0028
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8464	0.0268
	RS	FARC-HD	cpu	mips	bw	–	–	–	–	0.8725	0.0139
		FURIA	cpu	energy	mips	bw	mem	–	–	0.9880	0.0025
		IVTURS	cpu	storage	bw	energy	–	–	–	0.8648	0.1445
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8249	0.0165
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9252	0.0191
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.7799	0.0301
LR	MC	Chi	cpu	storage	mem	bw	mips	energy	–	0.9063	0.0164
		FARC-HD	cpu	bw	storage	mips	mem	–	–	0.9546	0.0082
		FURIA	cpu	energy	storage	mips	–	–	–	0.9916	0.0045
		IVTURS	cpu	bw	mips	mem	–	–	–	0.9644	0.0099
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.7105	0.1008
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9315	0.0080
	MMT	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9771	0.0079
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9227	0.0087
	MU	Chi	cpu	mips	energy	–	–	–	–	0.5852	0.0087
		FARC-HD	cpu	bw	energy	–	–	–	–	0.9261	0.0048
		FURIA	cpu	energy	storage	mem	–	–	–	0.9958	0.0013
		IVTURS	cpu	bw	mips	mem	storage	–	–	0.9492	0.0131
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.5796	0.0100
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9208	0.0093
LR	MMT	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9662	0.0111
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9015	0.0042
	MU	Chi	cpu	mips	mem	storage	energy	bw	–	0.6302	0.0121
		FARC-HD	cpu	bw	mips	–	–	–	–	0.8782	0.0206
		FURIA	mips	energy	cpu	mem	bw	–	–	0.9945	0.0018
		IVTURS	cpu	mem	bw	storage	energy	mips	–	0.8349	0.1493
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.5422	0.0039
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8533	0.0338
LR	MMT	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9484	0.0091
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.7621	0.1016
	MU	Chi	cpu	storage	bw	mem	mips	energy	–	0.9326	0.0045
		FARC-HD	cpu	bw	energy	–	–	–	–	0.9358	0.0080
		FURIA	mem	energy	cpu	mips	–	–	–	0.9894	0.0039
		IVTURS	cpu	bw	mips	mem	–	–	–	0.9403	0.0051
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8471	0.0087
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9261	0.0025
LR	MMT	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9778	0.0045
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9135	0.0074
	MU	Chi	cpu	storage	bw	mem	mips	energy	–	0.9326	0.0045
		FARC-HD	cpu	bw	energy	–	–	–	–	0.9358	0.0080
		FURIA	mem	energy	cpu	mips	–	–	–	0.9894	0.0039
		IVTURS	cpu	bw	mips	mem	–	–	–	0.9403	0.0051
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.8471	0.0087
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	–	0.9261	0.0025

5.2. Analyzing results from IQR and LR allocation algorithms

Next, we discuss the mean classification values of FARC-HD, Chi, FURIA, and IVTURS. In Table 4, “bw” represents the available bandwidth, “cpu” denotes the percentage of CPU utilization, “energy” indicates the power consumption, “mips” means the percentage usage of full potential, “ram” denotes the percentage of RAM usage and “storage” denotes the percentage of disk utilization.

Table 4 presents the analysis based on the average results, integrating the four Feature Selection algorithms with the IQR and LR allocation algorithms. See the highlighted text in the best search results.

Feature Selection (FS) compared with the Int-FLBCC configuration demonstrated superior performance across all scenarios, indicating enhanced classification capability and the potential for system refinement to address diverse scenarios. In both cases, the best results were obtained using the FURIA algorithm.

These results are related to the mean values obtained through simulations of the FURIA algorithm (FRBCS × Int-FLBCC), graphically depicted in Fig. 5. They reveal variations ranging from 0.15 (IQR×MU)

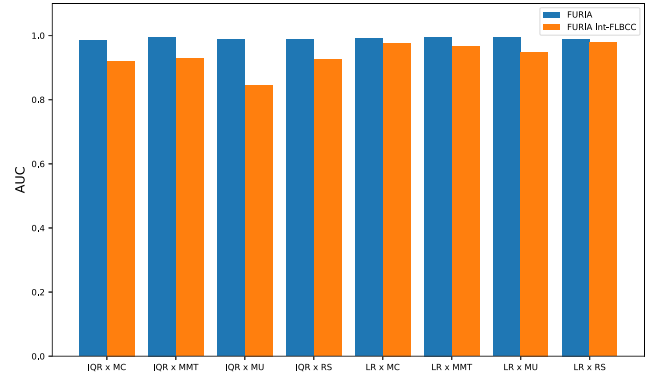


Fig. 5. Results of FURIA's Simulation (FRBCS and Int-FLBCC) for Feature Selection Performed on IQR and LR Policies.

to 0.01 (LR×RS), highlighting the algorithm's sensitivity to different input parameters and configurations.

During the execution of SFS, whenever the first iteration resulted in identical values for all variables, the run was discarded due to the impossibility of selecting the best performance. This occurs in the Chi algorithm when the allocation policies IQR or MAD are selected, as well as in the combination of LRR × MU. For FARC-HD and IVTURS, this happens in the combinations IQR × MU, MAD × MC, MAD × MU, MAD × RS, and the configuration generated by the Feature Selection in MAD × MMT.

The result obtained by the Feature Selection demonstrates that the “CPU” variable is more relevant than the other variables, given that it appears in all scenarios for both the IQR and LR policies. This happens for FURIA in the combinations: IQR×MU, LR×MU, and LR×RS, although it does not hold the top position.

As the primary variable, the CPU selection is compatible with the specialists' configuration in setting up the Int-FLBCC (“cpu”, “mem”, and “bw”).

About the second most prominent variable, the results for the IQR and LR policies differ slightly, where “bw” emerges as the second place for IQR and “mips” for LR. Additionally, in the third position, for IQR, “mips” is observed, while for LR, there is a tie between “bw” and “energy”. In these cases, there are differences when compared with the specialist's definitions. See that just “bw” is equal, in the third position in LR.

If we consider only the best results, obtained by FURIA, the number of most selected variables and their order change significantly from the definition in Int-FLBCC.

Fig. 6 depicts the number of variables selected by the Feature Selection under the IQR and LR policies, compared with the number of variables solely selected by FURIA in the same scenario.

To demonstrate the influence of variables' order in the performance of classification algorithms, Table 5 presents the classification performance results for all combinations of variables selected by SFS using FURIA for the IQR×MMT policies (“cpu”, “energy”, “mem”, “storage”). This table shows that the best combination would be “energy”, “cpu”, “storage”, and “mem”.

However, as we can observe in Fig. 7, the incremental methodology of SFS directed the best path to start with the “cpu” as the first variable, which achieved a performance close to 0.925, instead of the “energy” variable, whose performance was much lower, around 0.800.

So, in this study, we consider retaining the result obtained by the SFS methodology and discarding the best result obtained by permutation. This decision was based on the premise that data reproducibility justifies using a standardized method.

The performance evolution obtained by SFS ensures that the increment on each variable achieves the best results. During the Feature Selection, the performance results obtained for each combination of

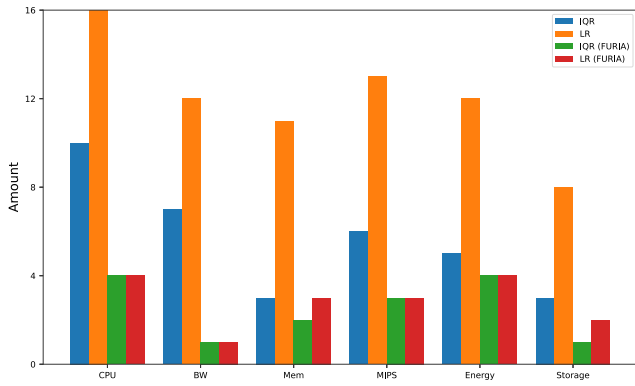


Fig. 6. Comparison of the Number of Selected Variables in the FS With the Execution of FURIA on IQR and LR Policies.

Table 5

Classification Performance Results Obtained by Permuting the Variables Selected by FURIA for the IQR x MMT Policies.

Variables				AUC
energy	cpu	storage	mem	0.9964
mem	energy	cpu	storage	0.9961
cpu	energy	storage	mem	0.9961
mem	storage	cpu	energy	0.9961
energy	mem	cpu	storage	0.9961
mem	cpu	storage	energy	0.9961
mem	energy	storage	cpu	0.9961
cpu	storage	energy	mem	0.9961
storage	energy	mem	cpu	0.9961
energy	cpu	mem	storage	0.9958
storage	energy	cpu	mem	0.9958
cpu	mem	storage	energy	0.9958
energy	storage	cpu	mem	0.9958
cpu	storage	mem	energy	0.9958
storage	mem	energy	cpu	0.9958
mem	storage	energy	cpu	0.9958
storage	mem	cpu	energy	0.9958
energy	mem	storage	cpu	0.9958
cpu	mem	energy	storage	0.9957
storage	cpu	mem	energy	0.9954
cpu	energy	mem	storage	0.9954
storage	cpu	energy	mem	0.9954
energy	storage	mem	cpu	0.9954
mem	cpu	energy	storage	0.9954

IQR and LR allocation policies, with the selection policies, considered the increment of one more variable whenever there was an increase in performance of at least 1%.

Considering the best performance results (FURIA), in addition to surpassing the performance of the configuration defined by the specialists, the FS brought modifications, including an increase in variables, as is the case for the scenarios IQR×MMT, IQR×RS, and all combinations of LR (see Fig. 8).

5.3. Analyzing results from LRR and MAD allocation algorithms

Table 6 presents the analysis based on the average results, integrating the four Feature Selection algorithms regarding the LRR and MAD allocation algorithms. See the best search results highlighted in the text.

Once again, Feature Selection demonstrated superior performance compared with the Int-FLBCC configuration across all scenarios, indicating enhanced classification capability and suggesting potential for system refinement to accommodate diverse scenarios. Notably, FURIA consistently outperformed other algorithms in all instances.

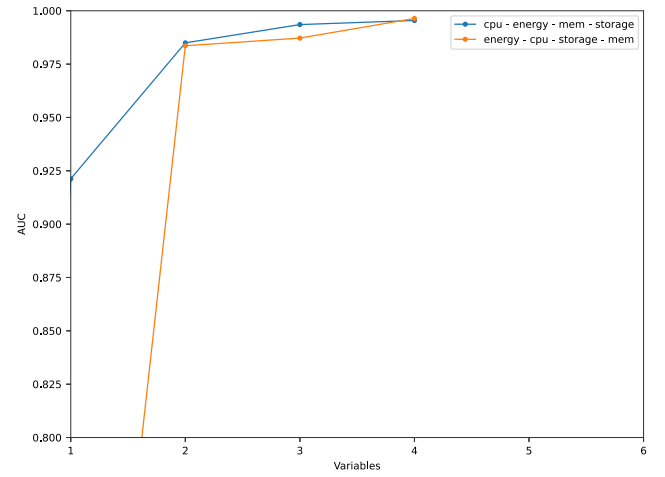


Fig. 7. Comparison of Performance Evolution between the Best Result from Permutation and Obtained by SFS with FURIA in the IQR x MMT Scenario.

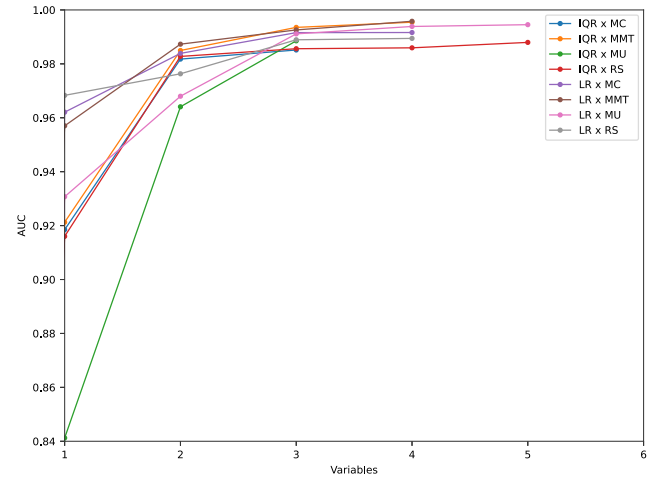


Fig. 8. FURIA's Simulation for Feature Selection Performed on IQR and LR Policies.

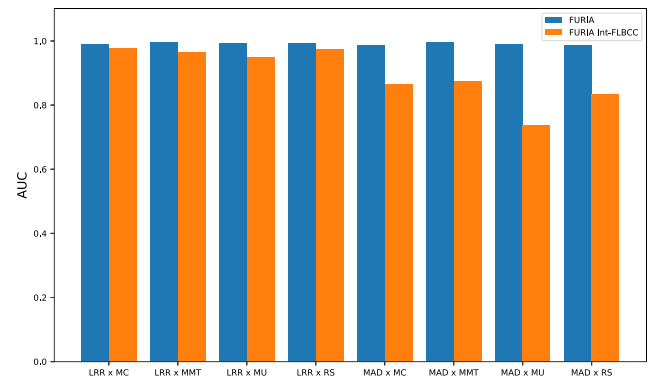


Fig. 9. FURIA's Simulation (FRBCS and Int-FLBCC) for Feature Selection Performed on LRR and MAD Policies.

The results refer to the mean values obtained through simulations using the FURIA algorithm (FRBCS × Int-FLBCC), as depicted in Fig. 9. These findings highlight variations ranging from 0.25 (MAD×MU) to 0.01 (LRR×MC), underscoring the algorithm's sensitivity to different input parameters and configurations.

Again, if the first iteration of SFS resulted in identical values for all variables, the run is discarded due to the impossibility of selecting the

Table 6

Global Results of Feature Selection Simulation for LRR and MAD.

AP	SP	FRBCS	Variables						AUC	stddev
LRR	MC	Chi	cpu	storage	mem	bw	mips	energy	0.9063	0.0164
		FARC-HD	cpu	bw	storage	mips	mem	–	0.9546	0.0082
		FURIA	cpu	energy	storage	mips	–	–	0.9916	0.0045
		IVTURS	cpu	bw	mips	mem	–	–	0.9644	0.0099
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	0.7105	0.1008
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9315	0.0080
	MMT	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9771	0.0079
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9227	0.0087
		Chi	cpu	mips	energy	–	–	–	0.5852	0.0087
		FARC-HD	cpu	bw	energy	–	–	–	0.9261	0.0048
		FURIA	cpu	energy	storage	mem	–	–	0.9958	0.0013
		IVTURS	cpu	bw	mips	mem	storage	–	0.9492	0.0131
MAD	MC	Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	0.5796	0.0100
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9208	0.0093
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9662	0.0111
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9015	0.0042
	MU	FARC-HD	cpu	bw	mips	–	–	–	0.8782	0.0206
		FURIA	mips	energy	cpu	mem	bw	–	0.9945	0.0018
		IVTURS	cpu	mem	bw	storage	energy	mips	0.8349	0.1493
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	0.5422	0.0039
		FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	0.8533	0.0338
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9484	0.0091
MAD	MMT	IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	0.7621	0.1016
		Chi	cpu	mem	bw	mips	energy	storage	0.8942	0.0057
		FARC-HD	cpu	bw	storage	–	–	–	0.9342	0.0102
		FURIA	cpu	energy	storage	–	–	–	0.9920	0.0029
		IVTURS	cpu	bw	mips	mem	storage	–	0.9595	0.0071
		Chi (Int-FLBCC)	cpu	mem	bw	–	–	–	0.6835	0.1261
	RS	FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9231	0.0155
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9744	0.0057
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	0.9054	0.0158
		FURIA	cpu	energy	mips	–	–	–	0.9870	0.0036
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.8648	0.0266
		IVTURS	cpu	energy	mem	storage	–	–	0.9952	0.0028
MAD	MMT	FARC-HD (Int-FLBCC)	cpu	mem	bw	–	–	–	0.5694	0.0852
		FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.8735	0.0234
		IVTURS (Int-FLBCC)	cpu	mem	bw	–	–	–	0.5649	0.0796
		FURIA	cpu	energy	mem	bw	–	–	0.9906	0.0032
	MU	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.7362	0.0144
		FURIA	mips	energy	cpu	mem	storage	–	0.9873	0.0047
	RS	FURIA (Int-FLBCC)	cpu	mem	bw	–	–	–	0.8339	0.0307

best performance. For LRR and MAD, this occurs in the Chi algorithm when the allocation policy is MAD and in the combination of LRR×MU. For FARC-HD and IVTURS, this happens in the combinations MAD×MC, MAD×MU, MAD×RS, and the configuration generated by the Feature Selection in MAD×MMT.

The result obtained by the FS demonstrates that the variable “cpu” is more relevant than the other variables, appearing in all scenarios for both the LRR and MAD policies, even though it does not hold the top position, which happens for the FURIA in the combinations LRR×MU and MAD×RS.

The selection of “cpu” as the primary variable aligns with the configuration defined by the specialist in setting up the Int-FLBCC (“cpu”, “mem”, and “bw”).

Concerning the second most prominent variable, the results differ significantly between the LRR and MAD policies. In LRR, there is a tie between “bw” and “mips” for the second and third places, while in MAD, a tie for the first place between “cpu” and “energy” results in a shared second place, with “mem” appearing in the third place.

If we consider only the best results, again obtained by FURIA, the number of the most selected variables and their order change significantly from the variable definition adopted in Int-FLBCC.

Fig. 10 expresses the number of variables selected by the FS under the LRR and MAD policies, compared with the number of variables selected solely by FURIA, in the same scenarios.

Once again, the variables’ order significantly influences the classification algorithms’ performance. Nonetheless, we rely on values obtained through the SFS methodology as a standardized approach.

The performance evolution obtained by SFS ensures the best result for each variable increment. During the FS, the performance results for each combination of LRR and MAD allocation policies with the selection policies considered, adding one more variable whenever there was an increase in performance of at least 1%. Fig. 11 presents the performance evolution for LRR and MAD.

Considering the best performance results (FURIA), in addition to surpassing the performance of the configuration defined by the specialist, the FS brought a modification to this configuration, including

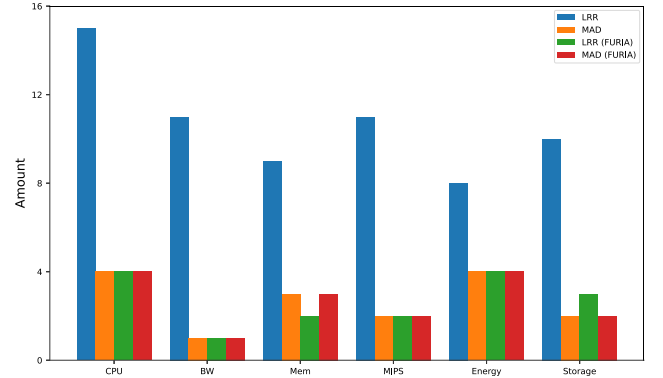


Fig. 10. Comparison of the Number of Selected Variables in the FS and the FURIA's Execution on LRR and MAD Policies.

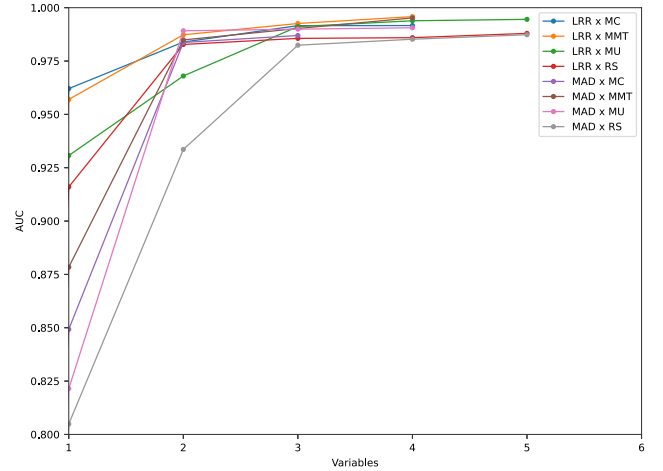


Fig. 11. Result Details of FURIA's Simulation for Feature Selection Performed on LRR and MAD Policies.

an increase in variables, as is the case for the LRR × MC, LRR × MMT, LRR × MU, MAD × MMT, MAD × MU, and MAD × RS scenarios.

5.4. Int-FLBCC improvement

Based on the results obtained from the Feature Selection, we propose an adaptation of the Int-FLBCC adjustments to different allocation combinations and selection policies. In this way, new configurations for the input variables of the fuzzy system, integrating new rule base definitions specific to each scenario, can improve the system with an enhancement in the classification performance of physical machines (hosts) utilization level for the VM consolidation task.

Fig. 12 presents the Hybrid Fuzzy Load Balancing for Cloud Computing (Hybrid-FLBCC), the proposed adaptation of Int-FLBCC. The system does not consider only “cpu”, “mem”, and “bw” as input variables. In addition to this standard configuration, the system allows for selecting a dynamic configuration based on the combination of defined allocation and selection policies. Following the same method, we can apply the rule base to a specific set of rules loaded for each scenario.

The definition of new rule bases and membership functions will utilize the knowledge generated by the FRBCS.

The results obtained by FURIA outperformed other algorithms in all scenarios. However, being a Type-1 Fuzzy approach, the results do not capture the levels of uncertainty and imprecision inherent in Cloud Computing. On the other hand, IVTURS, by addressing Type-2 Fuzzy Sets, encompasses these requirements. Therefore, to pursue FURIA's performance while meeting the environment's requirements,

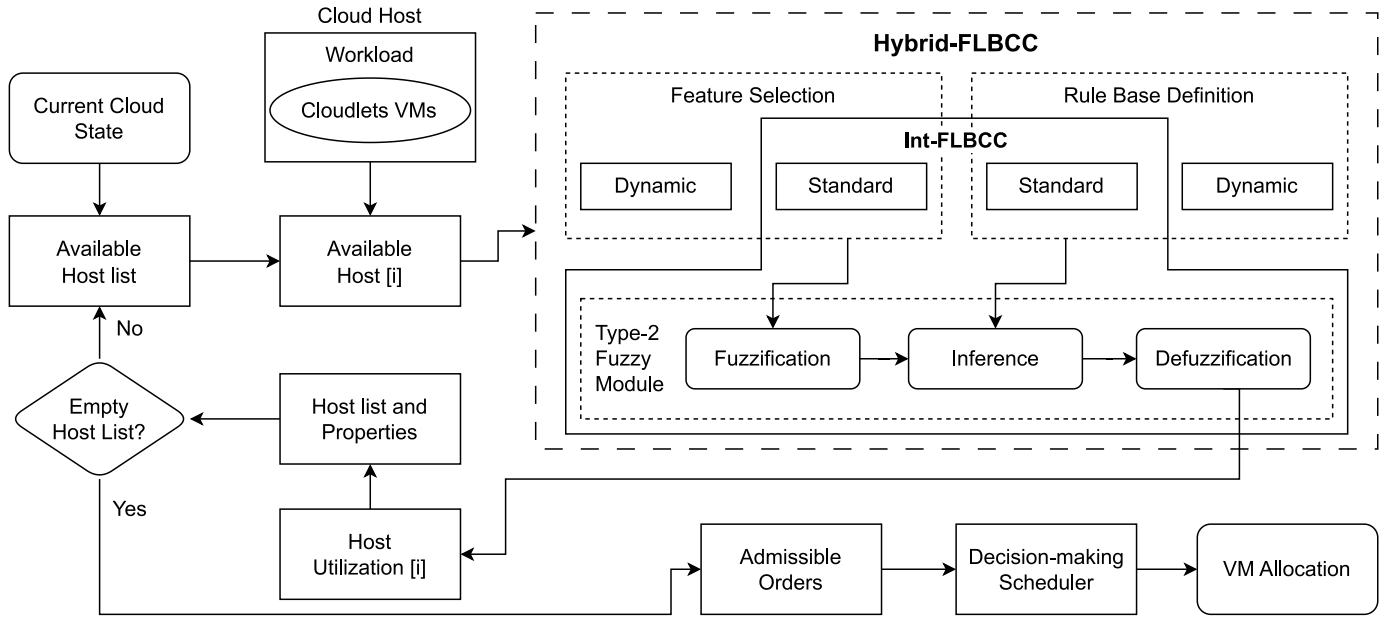


Fig. 12. Overview of the Hybrid-FLBCC Extending the Int-FLBCC Approach.

characteristics of IVTURS configurations will be reviewed to integrate efficiency and imprecision modeling.

Moreover, future enhancements may also explore the integration of hybrid neuro-fuzzy models to improve the adaptability and effectiveness of the decision-making process in heterogeneous and dynamic environments. These approaches have demonstrated promising results in related contexts, such as task runtime prediction in edge computing scenarios [51], and may contribute to the evolution of the Hybrid-FLBCC strategy by enabling more intelligent and context-aware scheduling mechanisms.

6. Conclusions

This article explores the Feature Selection methodology to highlight the most relevant variables for server consolidation as an extension proposal of Int-FLBCC. Thus, this strategy facilitates the definition of new membership functions, improving the rule sets and providing a straightforward interpretation. Moreover, this optimized approach simplifies the fuzzy system implementation and may lead to saving computational resources.

The objective was to propose a new set of variables better suited to the various execution scenarios provided by CloudSim, aiming to simplify fuzzy system simulation utilizing the computational intelligence of FRBCSs.

The comparative evaluation of classification algorithms revealed differences in their performances and approaches. FURIA stood out, demonstrating notable precision with consistently high AUC values, consolidating its effectiveness in classification. IVTURS showed a performance aligned with FURIA, positioning it as a promising alternative.

In all scenarios, comparing the results obtained from Feature Selection with those of the Int-FLBCC configuration demonstrated superior performance, indicating enhanced classification capability. Consequently, there is potential for system refinement by adapting it to address diverse scenarios.

6.1. Future research directions

While our study contributes to server consolidation, several promising avenues for future research emerge. Developing an interval-valued

fuzzy system for dynamic server consolidation aims to optimize energy consumption in cloud environments while maintaining a satisfactory SLA.

Our study advances the field of server consolidation using Fuzzy Logic, paving the way for future research. We propose extending the Int-FLBCC system, promoting a flexible approach, by enhancing the set of variables based on the outcomes of our Feature Selection process. To refine the inference modeling, adjust membership functions, and establish rule bases, we use the computational intelligence provided by FRBCSs.

The forthcoming stage of this study involves the development of the Hybrid-FLBCC approach, conceived as an evolutionary extension of the Int-FLBCC. This new strategy will leverage the concept of Flexible Computing to further enhance server consolidation efficiency in Cloud Computing environments.

The central idea is to exploit the computational intelligence of Fuzzy Rule-Based Classification Systems for the dynamic generation of the rule base and membership functions, thereby refining the inference process across various allocation combinations and selection policies.

By enabling the system to adopt new configurations for the input variables of the fuzzy system and to integrate scenario-specific rule bases, this approach improves the classification performance related to host utilization levels in VM consolidation tasks.

The implementation of the Hybrid-FLBCC approach will continue to rely on the CloudSim simulation environment, within which the fuzzy system will be integrated as a modular component. This module will act as an intermediary between the environmental scenario data and the scheduler's decision-making mechanism.

Subsequently, simulations will incorporate other workload traces from the PlanetLab project, enabling the comparison of energy efficiency performance between the Int-FLBCC and Hybrid-FLBCC strategies for host utilization level identification.

6.2. Research limitations

This study is based on several assumptions that guide both the design of the methodology and the interpretation of the results.

First, it assumes that the simulated cloud environment, implemented using CloudSim, adequately reflects the behavior of real-world

infrastructures, particularly regarding resource utilization dynamics and VM allocation patterns.

While simulation offers a controlled and reproducible setting for experimentation, it may not fully capture the complexity, unpredictability, and heterogeneity of production environments.

Additionally, the Feature Selection process assumes that the relationships among the selected variables remain stable across different policy combinations and workloads.

These limitations do not undermine the validity of the findings but indicate areas for refinement in future work, including validations with real datasets and integration with other uncertainty-handling strategies.

CRedit authorship contribution statement

R.R. Bastos: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **B.M.P. Moura:** Writing – review & editing, Supervision, Funding acquisition. **H.S. Santos:** Writing – review & editing, Supervision. **G. Lucca:** Writing – review & editing, Supervision, Funding acquisition. **A.C. Yamin:** Writing – review & editing, Supervision. **R.H.S. Reiser:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rafael Rodrigues Bastos reports financial support was provided by Coordination of Superior Level Staff Improvement. Bruno Moura Paz de Moura, Giancarlo Lucca, and Renata Hax Sander Reiser reports financial support was provided by Foundation for Research Support of Rio Grande do Sul State. Giancarlo Lucca and Renata Hax Sander Reiser reports financial support was provided by National Council for Scientific and Technological Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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